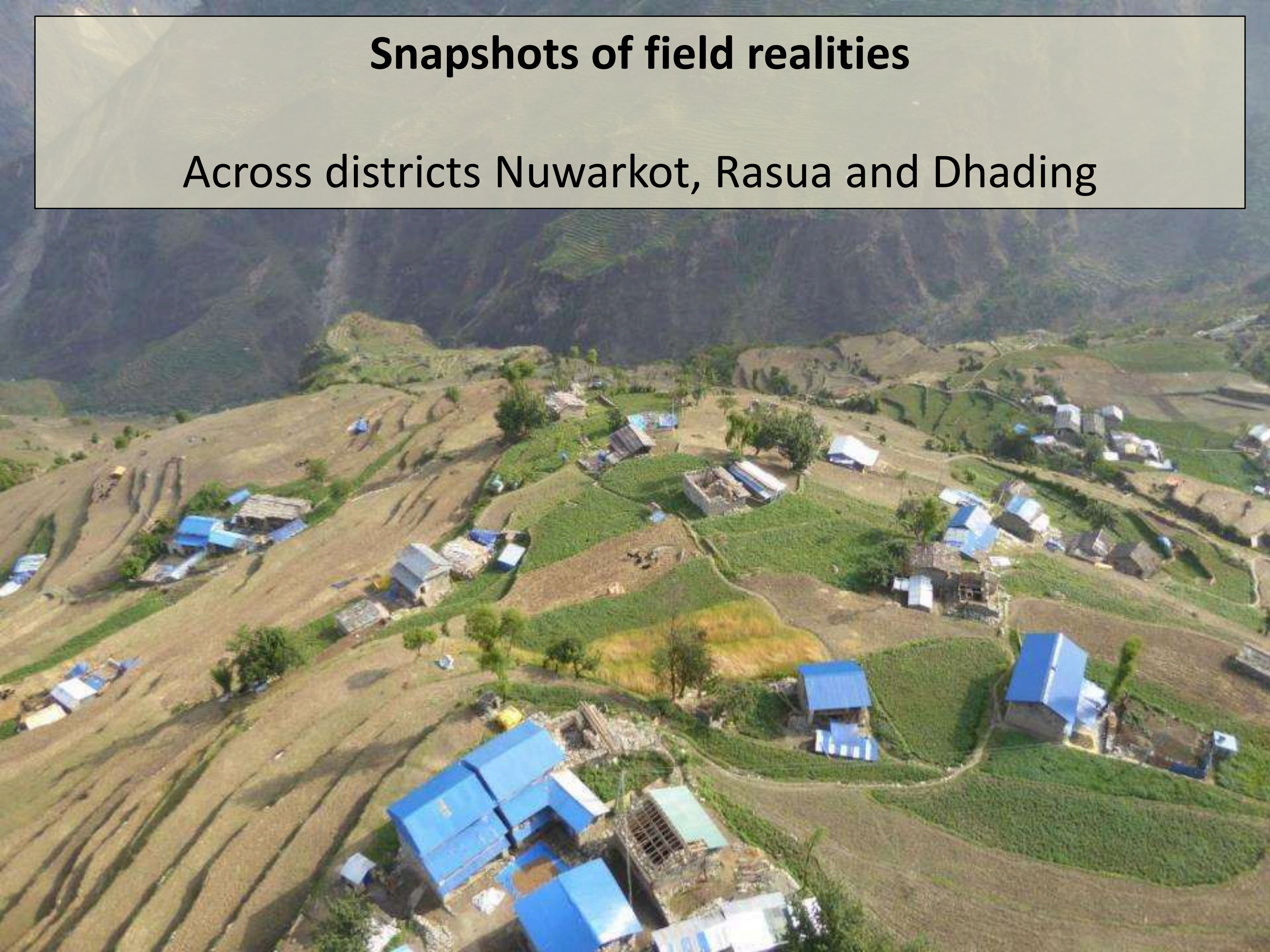
Snapshots of field realities

Across districts Nuwarkot, Rasua and Dhading



Who owns the land?

Many cases of title deeds belonging to a religious entity or a representative.

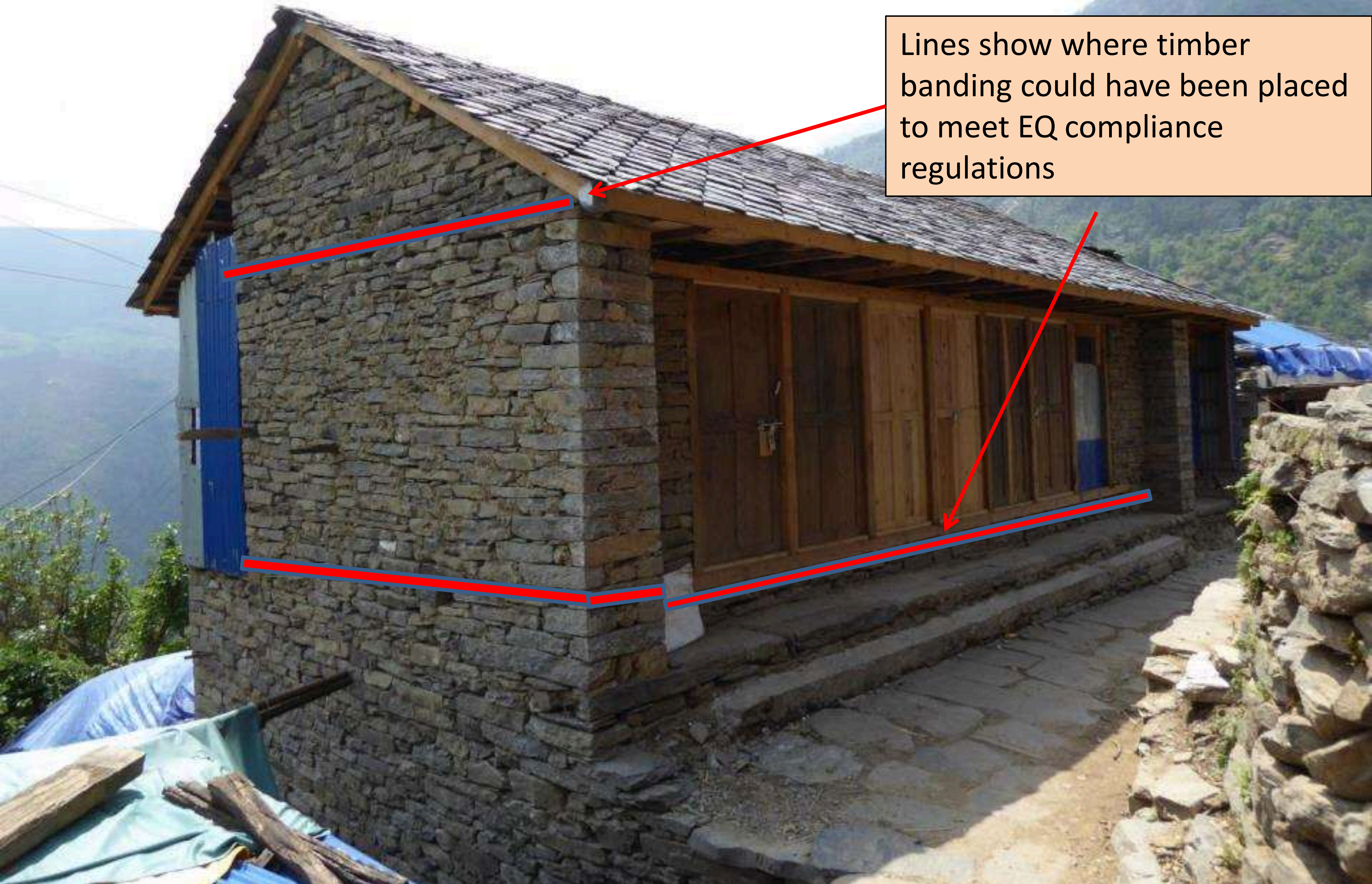
Will need quick resolution through the grievance process.

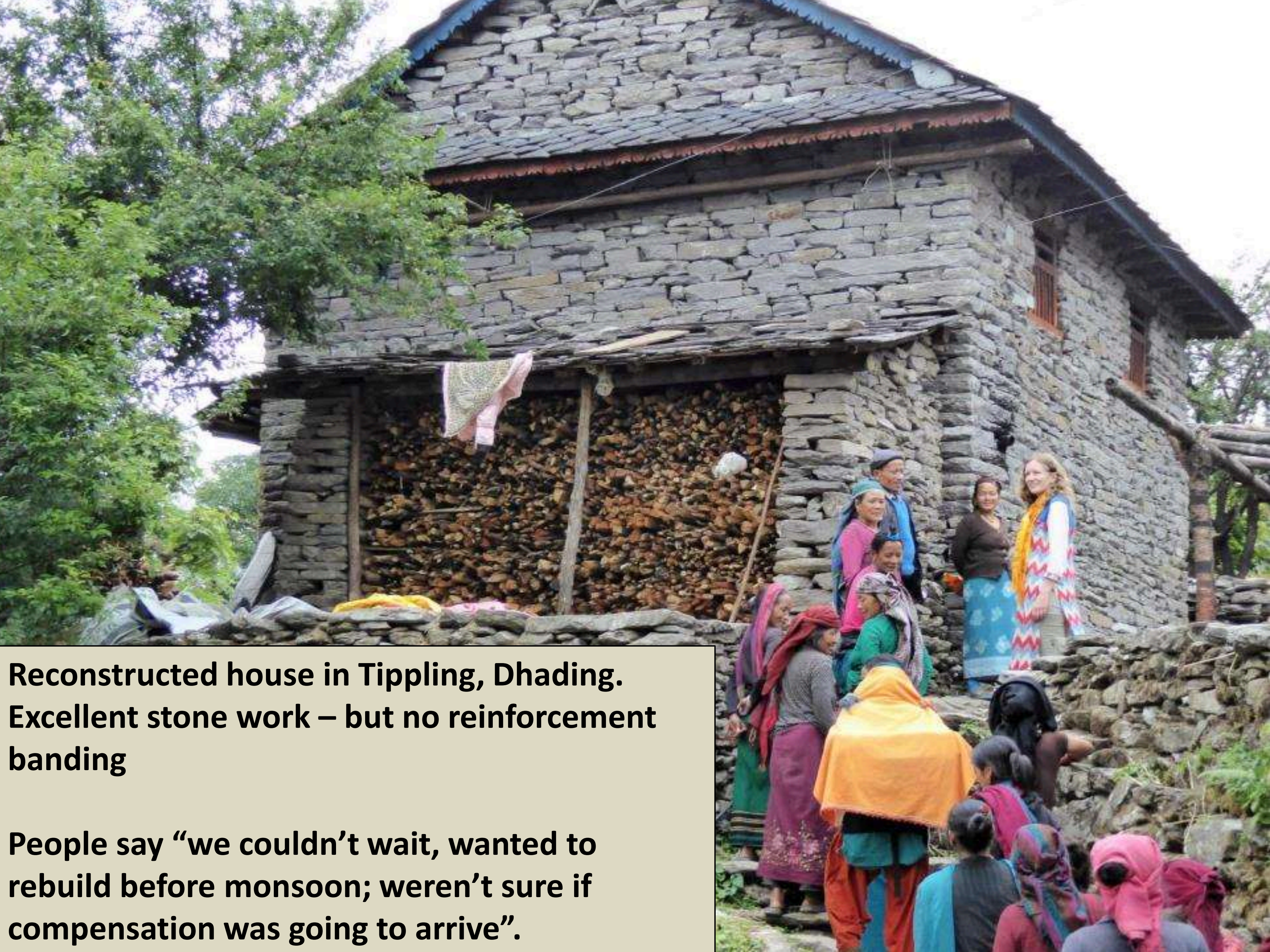
Listening to people's perspectives:

- How radio provides information on enrolment and compensation
- Housing important – but water is vital
- Without schools we need to send our children away to study

Reconstructed home in Haku (Rasua). Highly vulnerable to future earthquakes or tremors. Speeding up deliver of training and guidance is critical in such areas.

- North Dhading – Lapa VDC. Up to 70% reconstruction already
- EQ compliance (building code) not followed.
- Options for retrofitting?





Reconstructed house in Tippling, Dhading. Excellent stone work – but no reinforcement banding

People say "we couldn't wait, wanted to rebuild before monsoon; weren't sure if compensation was going to arrive".

Another building in Dhading under reconstruction – no guidance or training organisation has visited. Posters on the 10 key messages not available either.

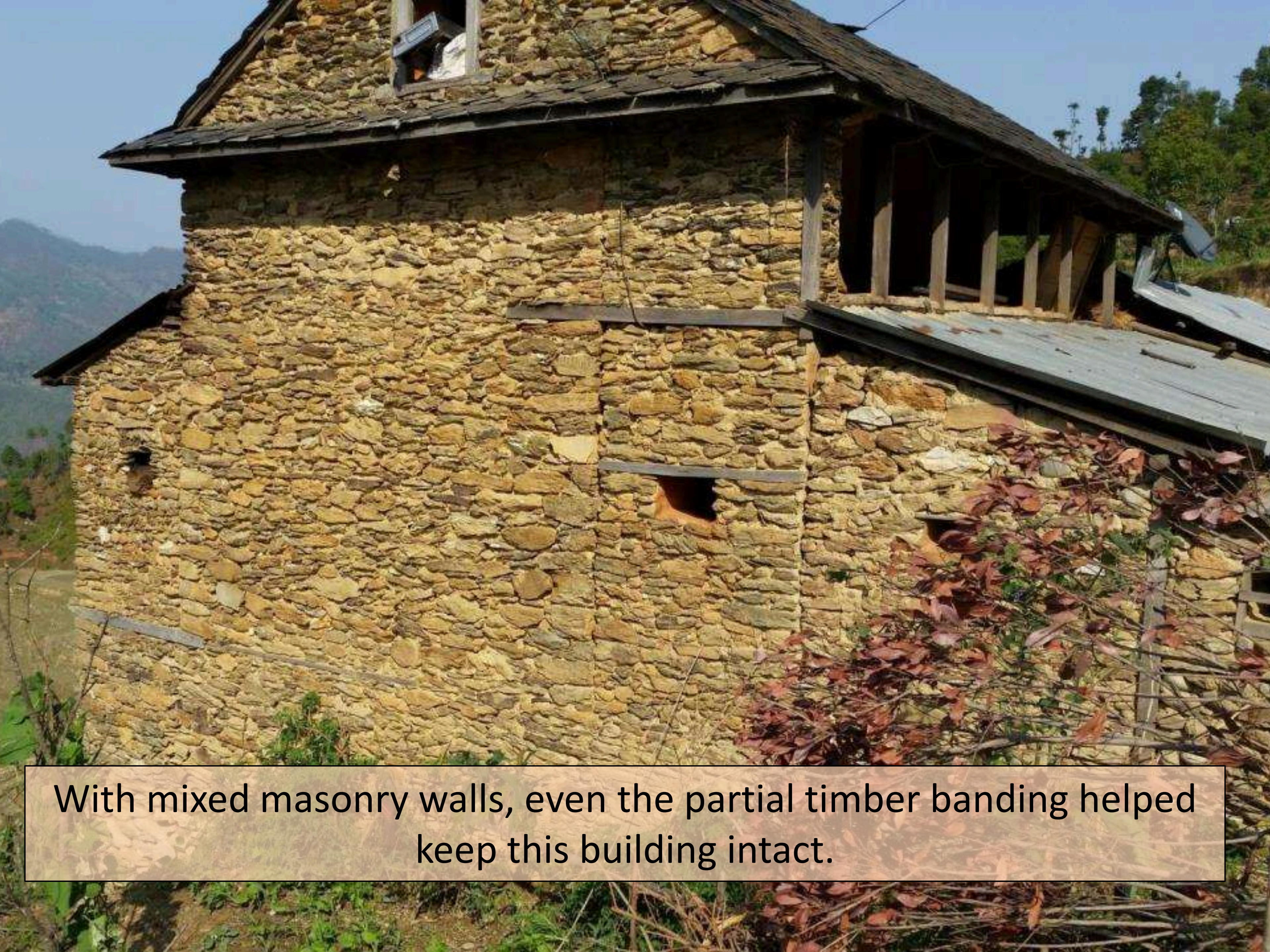
Local Timber is available
Lateral bands using timber would have been possible.

Up to 150,000 houses may have been rebuilt already.
What strategy on retrofitting / adding earthquake compliance after construction?

Haku, Rasua District. People keen to reconstruct before monsoon..
GOAL and OM Nepal recently approved to start training.

Buildings that withstood the earthquake

- Learning from good practice traditional architecture
- Well made timber banding around the house



With mixed masonry walls, even the partial timber banding helped keep this building intact.

Model house built in training provided by HELVATAS – showing good use of wooden bands.

Source: Dave Hodgkin, IOM / Shelter Cluster, 2015

Same village, Sindlepalchokc

Not because of they were not steel & concrete

Success was primarily due to good construction

1. Good Cornerstones
2. Frequent tie-stones
3. Reinforced gables
4. Strong banding
5. Rectilinear stones
6. Not too much mud
7. Floors tied to walls
8. Stable ground
9. Roof tied down to walls
10. Not too tall

Banding

- End eave as band
- Mid floor as band
- Strong foundation as band

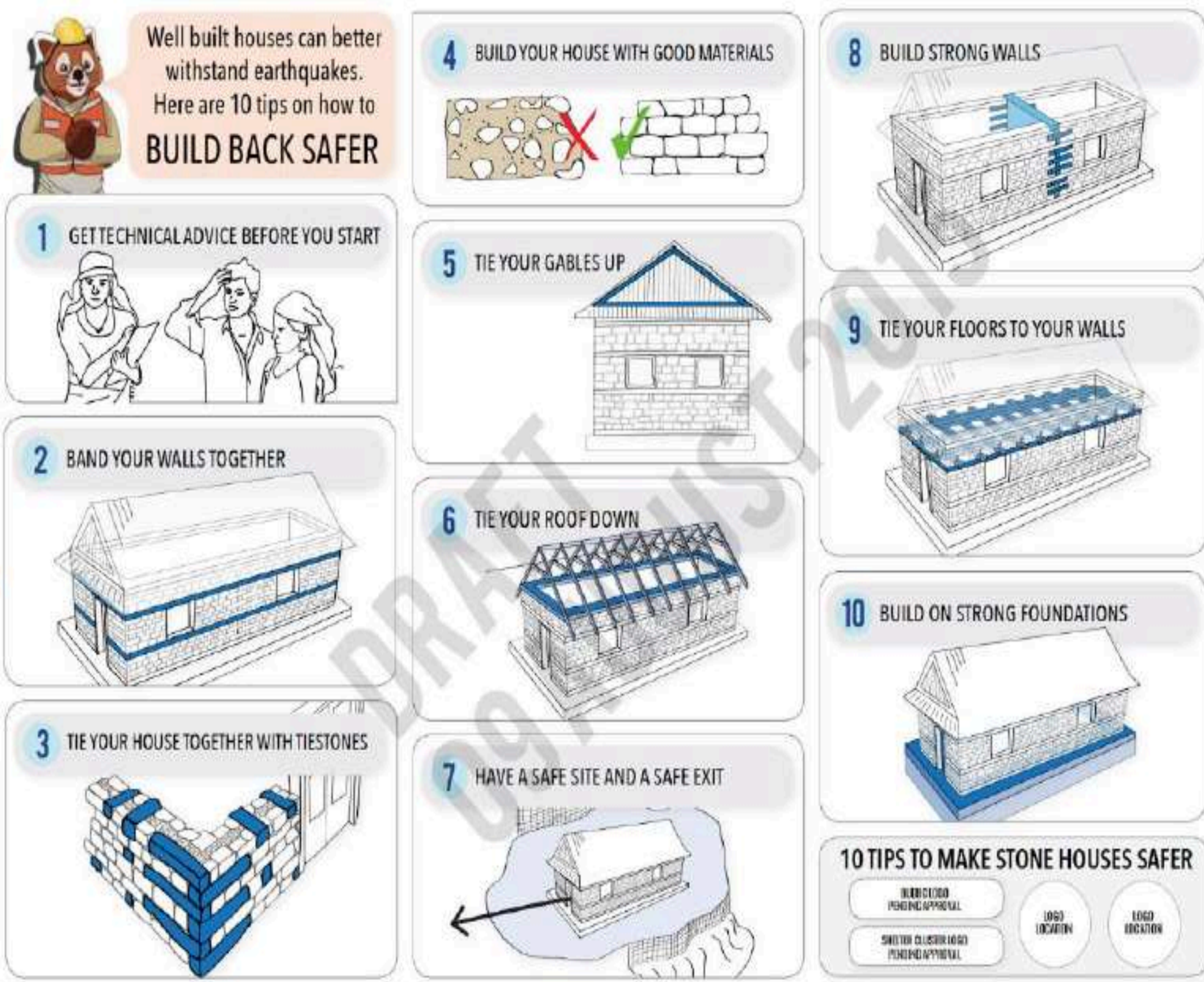
Veranda as Banding

Size/height

Building back properly IS the answer

10 Key Messages

IOM is working closely with the Shelter Cluster to develop 10 simple messages on how to build back safer in stone and mud



IOM • OIM

Haku – reconstruction without observation to safety measures (e.g. wooden banding).

Can this family still apply for housing compensation?


Earthbag building in Nuwarkot. Interesting low cost EQ-resistant system
- Only relevant where traditional stone not available
- Samaritan's Purse / Steadfast Nepal

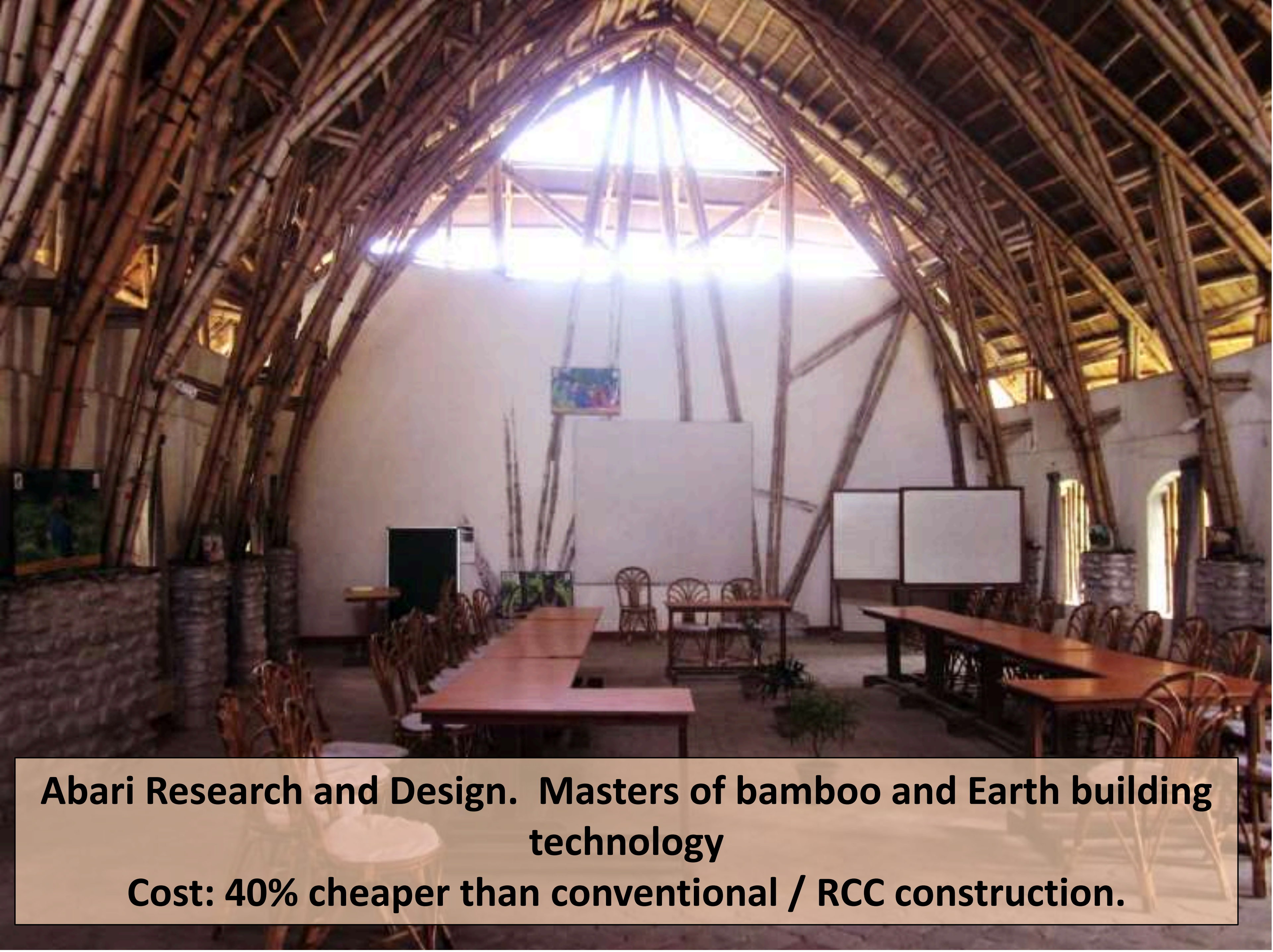
Mixed masonry with mud: local materials

= vulnerable housing

Buttress walls and well-tied walls – roof – corners = EQ compliance

Reinforced iron bars around each corner

Provides additional stability to prevent corner slippage in earthquakes



Abari Research and Design. Masters of bamboo and Earth building technology

Cost: 40% cheaper than conventional / RCC construction.

Abari’s bamboo vision

Transforming the landscape

Providing the community with seedlings

Collected more than 24 species of bamboo

Our objectives:

Bamboo plantation of 10,000 plants this year

A million bamboo plants in 10 years!

Combines livelihoods – landscape stability – building materials - environment & climate adaptation

- Internal plaster with earth
- Metal floor, framing and roof: about 30% cost of wood.
- Total cost: approx . 4.5 Lek, including labour.
- Without labour cost: 2.5 Lek

Building Materials – Wood

A critical supply chain issue

High potential for inflation

Isolated communities: local options only

Traditional methods of preparation are most common, obviously

Research needed on constraints to supply per region, options / alternatives

Langtang – Samaritans Purse imported this portable saw mill

Especially relevant for Langtang context

Whole forest destroyed by the landslide

Cutting ~ 200 cubic feet per day – enough for just over 1 house timber requirement per day. Great support from Nepal Army for logistics.

CBO Tippling – inspired and motivated – enormous resources! Recovery is an opportunity

Livelihoods and food security

Natural Resource management ?

- **Soil depletion, landslide risk**

- **Food insecurity, limited incomes;**

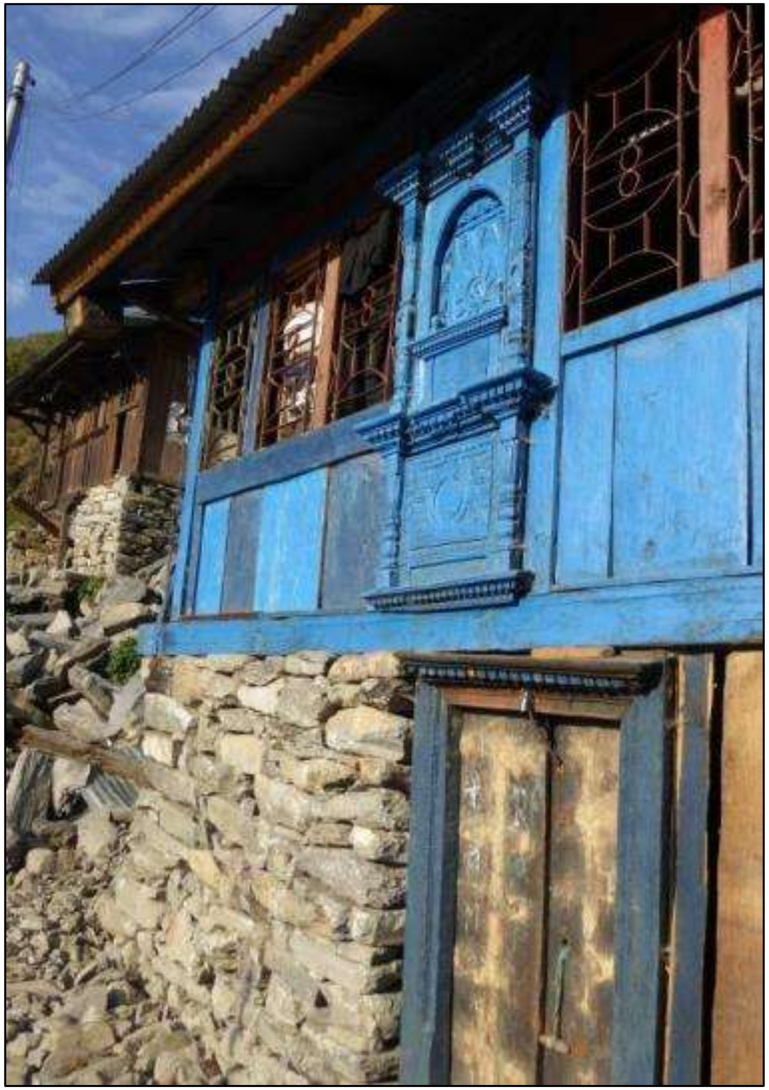
- **Insufficient income and food.**

- **Potential to stabilise slopes and increase biodiversity & incomes**

Much scope for improving agriculture techniques and systems to enhance productivity, address food insecurity, increase incomes. Need co-ordinated approach between Govt., donors, NGOs.

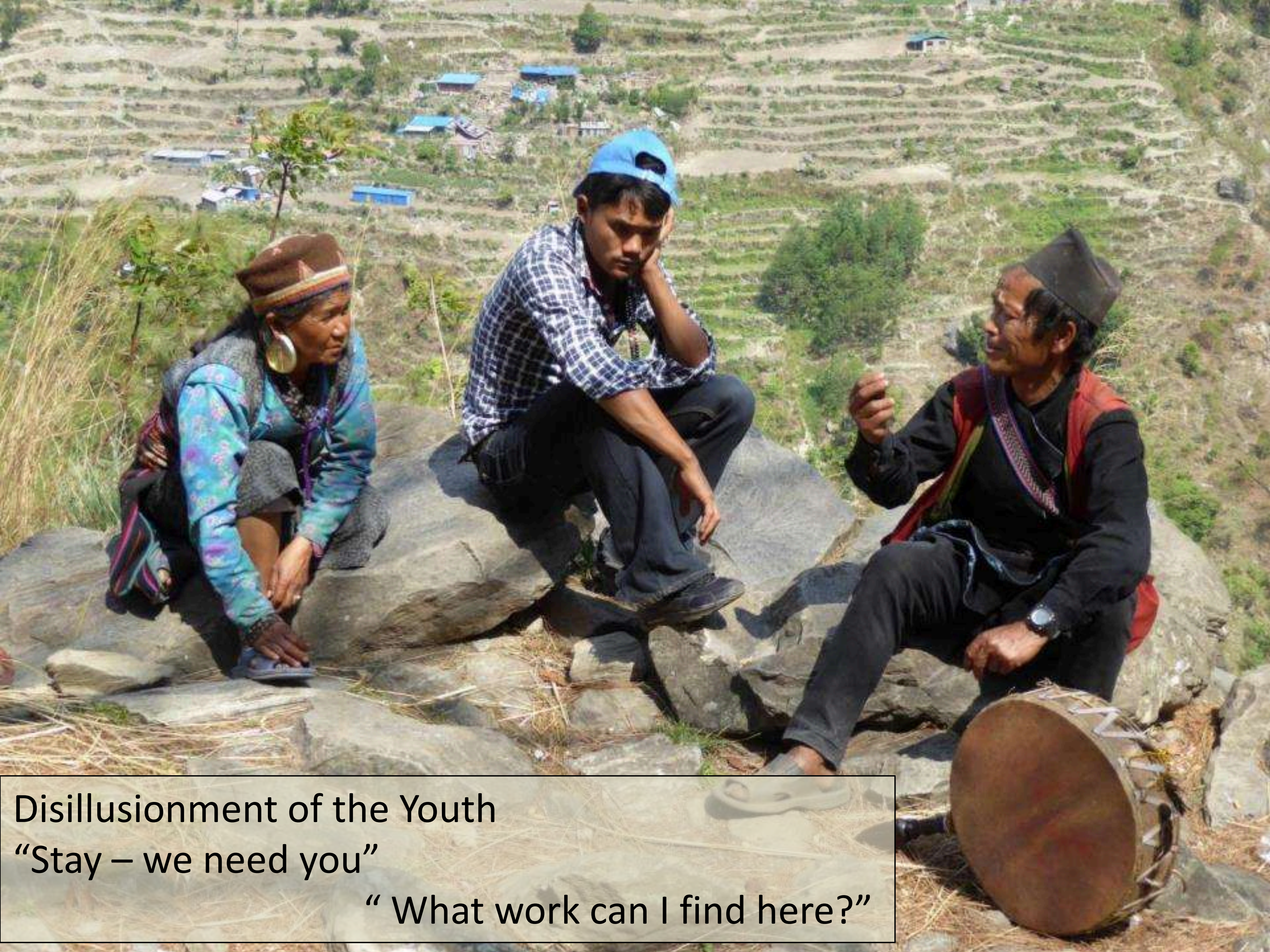
How can we integrate training in more sustainable agriculture into the reconstruction phase?

Migration labour (to Gulf, etc.) only other viable option to earn money.

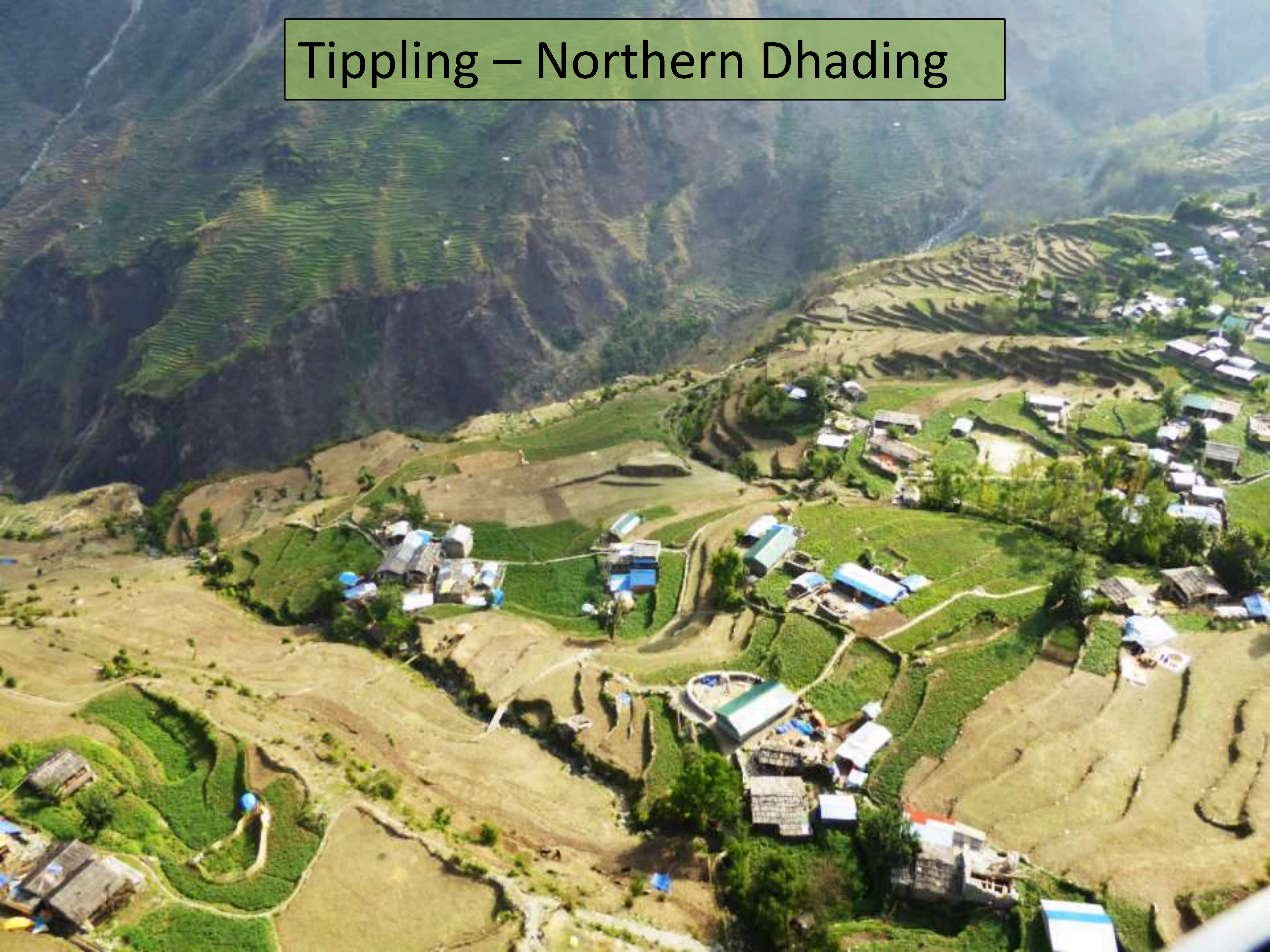



Opportunities to support local handicrafts and livelihoods throughout the recovery process.

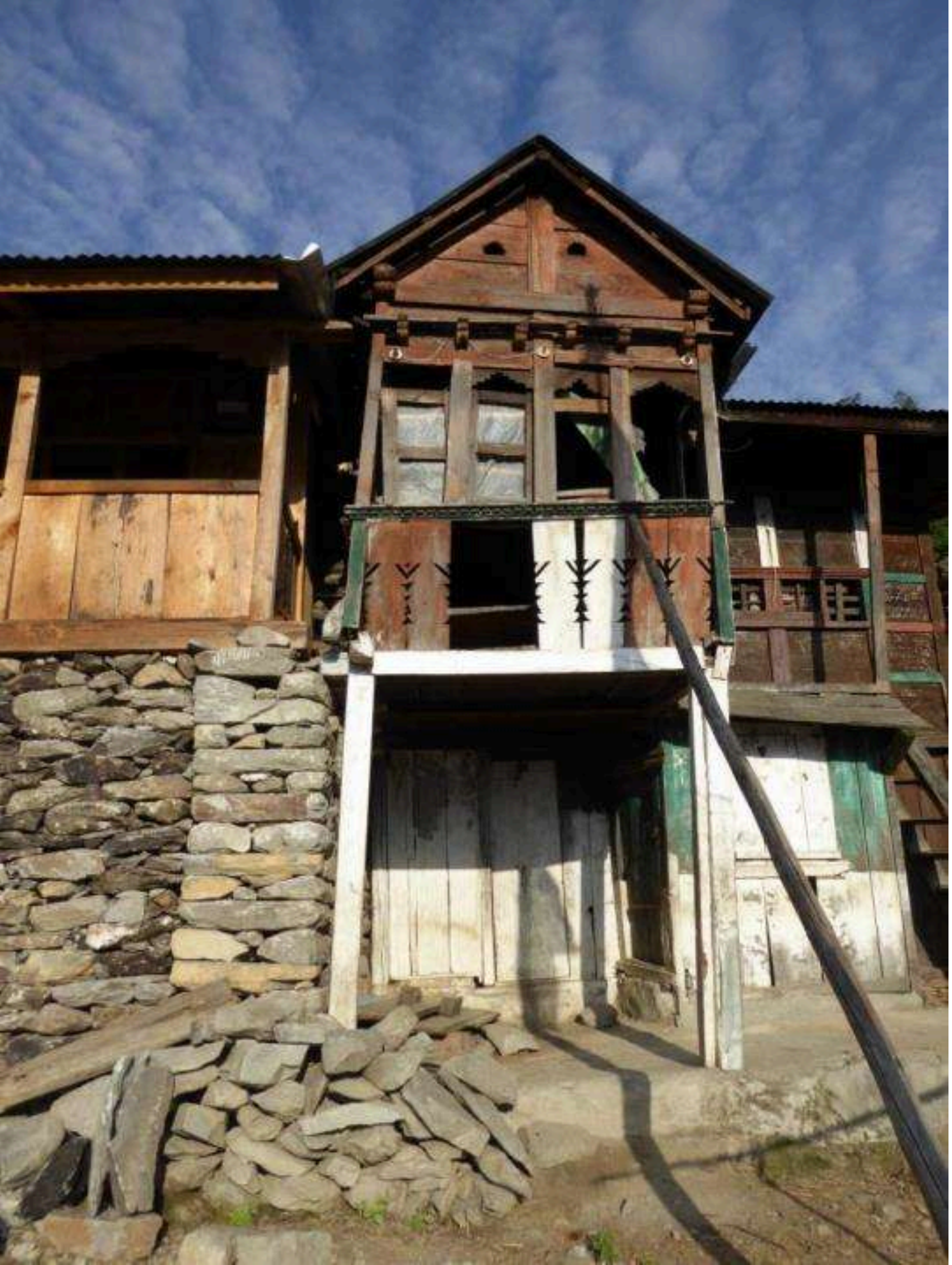
Can we encourage tourism?


Disillusionment of the Youth
“Stay – we need you”
“ What work can I find here?”

Tippling – Northern Dhading



Water supply – an urgent need everywhere we visited



Critical to maintain local cultural heritage in reconstruction

Cultural identity defined by vernacular architecture

Potential for tourist attraction and new livelihoods opportunities

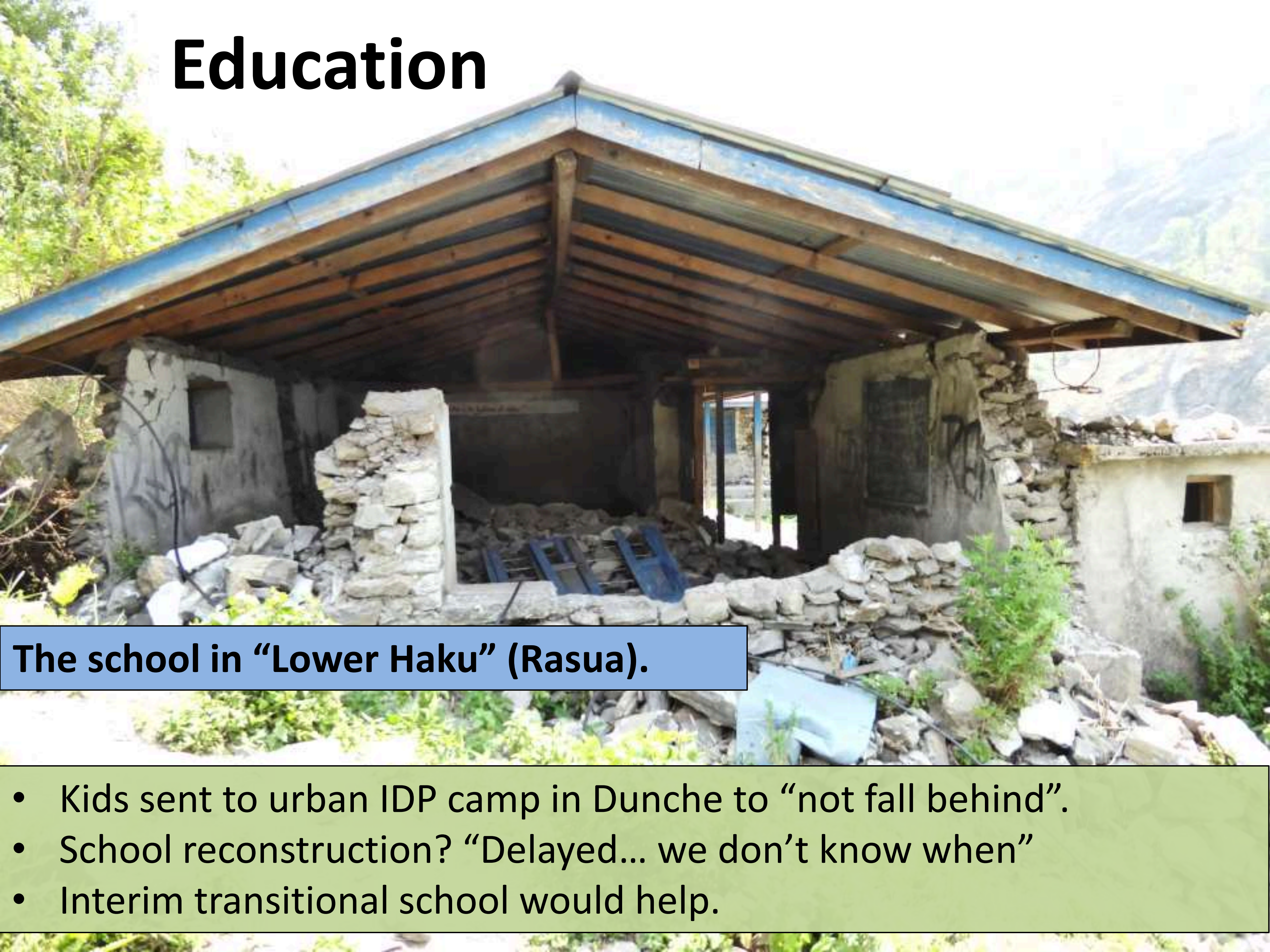
Social media / private sector to reach potential visitors for home stays and tourism?

Geological Surveys: need to communicate the plan of action with districts and communities. Which areas are no longer safe? What choices to people have?

Case study – Haku (Rasua district)
The people who survived, but can't go home

- In IDP camps – Nuwarkhot
- Awaiting resettlement options
- NRA / PDRF has specific allocation
- Information and communication!

Education



The school in "Lower Haku" (Rasua).

- Kids sent to urban IDP camp in Dunche to "not fall behind".
- School reconstruction? "Delayed… we don't know when"
- Interim transitional school would help.

Temporary school in tents, Rataon, N. Dhading. Many communities still have no temporary learning spaces – And this prevents return of IDPs in some areas.

Need to map out exactly which communities have and don't have temporary schools across each district, VDC and ward.

Extensive community consultations – Tippling

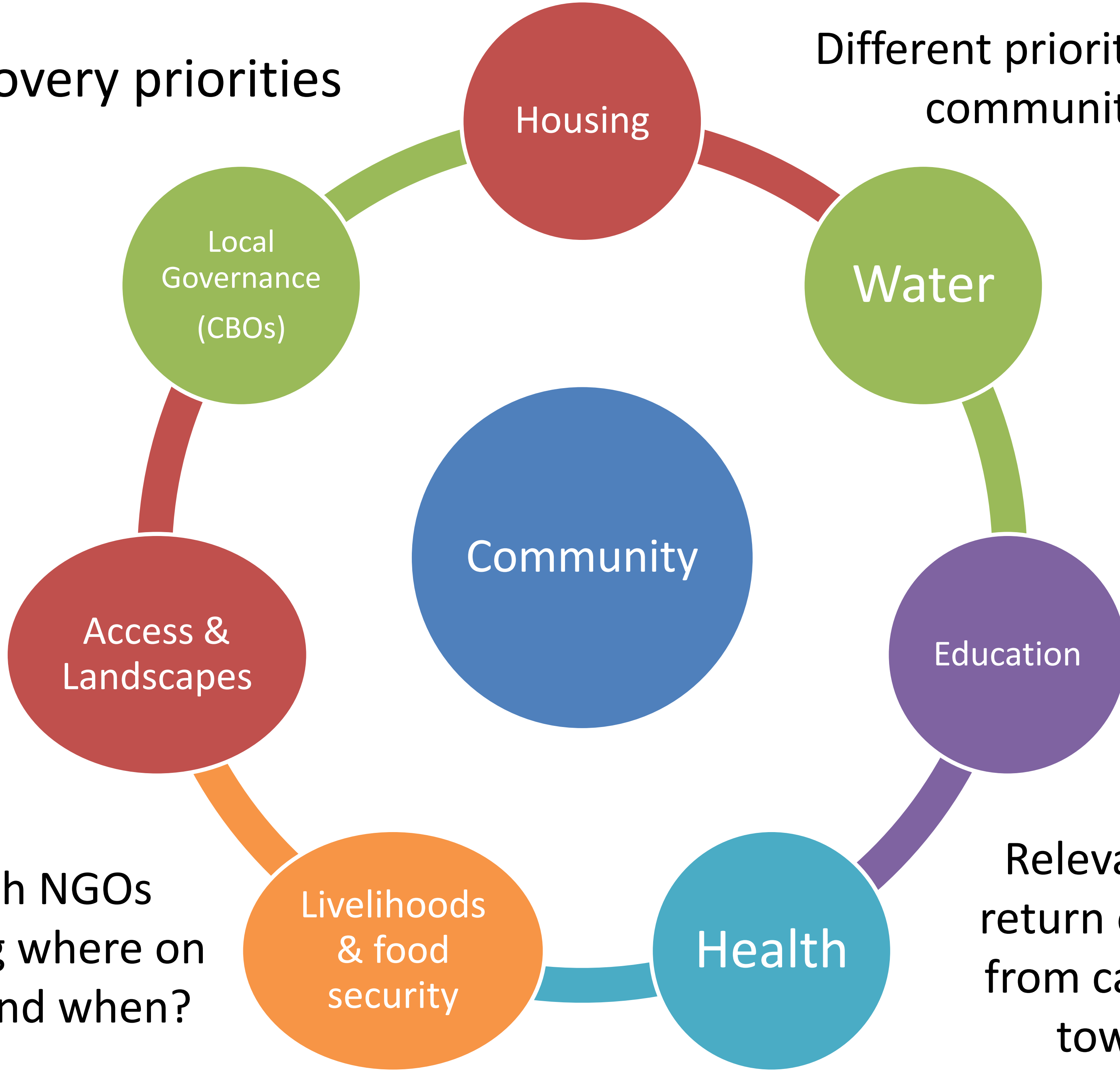
IDPs

- Camp coordination and management services
- Needs ongoing close coordination and collaboration with district authorities
- And MOHA / MOFALD
- Potential for NGO partnership combining CCCM and community infrastructure investments

Options and solutions for IDPs.
We need better integration between camp management
&
Return planning, restoration of basic services
&
Resettlement for those who cannot go home

Recovery priorities

Different priorities per community



Which NGOs working where on what and when?

Relevant to return of IDPs from camps / towns